

SLOME

Romance in Video Games

Technical & Design Handoff

Emotional NPC architecture, psychological foundations, implementation audit, and prioritized roadmap

Prepared for the incoming project group

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PROJECT CONTEXT

Cal Poly SURP — Romance in Videogames

ENGINE

Unity 6000.1.11f1

SOURCE SNAPSHOT

Uploaded Unity project and one-page project poster

HANDOFF DATE

5 August 2026

Core premise

The project treats romance as an evolving interaction between immediate emotion, relationship history, personality, and context—not as a single affection meter or a sequence of gifts that purchases a desired outcome.

How to Use This Handoff

This is both a design rationale and a technical audit. It explains what the team was trying to simulate, how the psychological models were translated into game systems, what the current Unity code actually does, and what the next group should fix before extending the prototype.

IMPORTANT SOURCE LIMITATION

This report is grounded in the uploaded Unity snapshot, the uploaded project poster, and retained project context from prior conversations available in this workspace. It does not claim access to every historical chat transcript. Where an intended design differs from the code, the report labels the distinction explicitly.

Document map

Section	What it answers
1	Executive summary and current truth
2	Project purpose and design motivations
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Source hierarchy

- **Code is the implementation truth.** When comments, the poster, and runtime wiring disagree, this report describes the code behavior first.
- **The poster is the concise design statement.** Its page-one diagram establishes the intended flow from player action through intent, PAD, relationship amplification, utility scoring, and NPC execution.
- **Psychological theory supplies vocabulary and structure, not proof of human equivalence.** PAD, Sternberg, and personality frameworks are used as abstractions for authoring and computation.
- **Recommendations are clearly separated.** Anything not present in the snapshot is described as future work rather than current functionality.

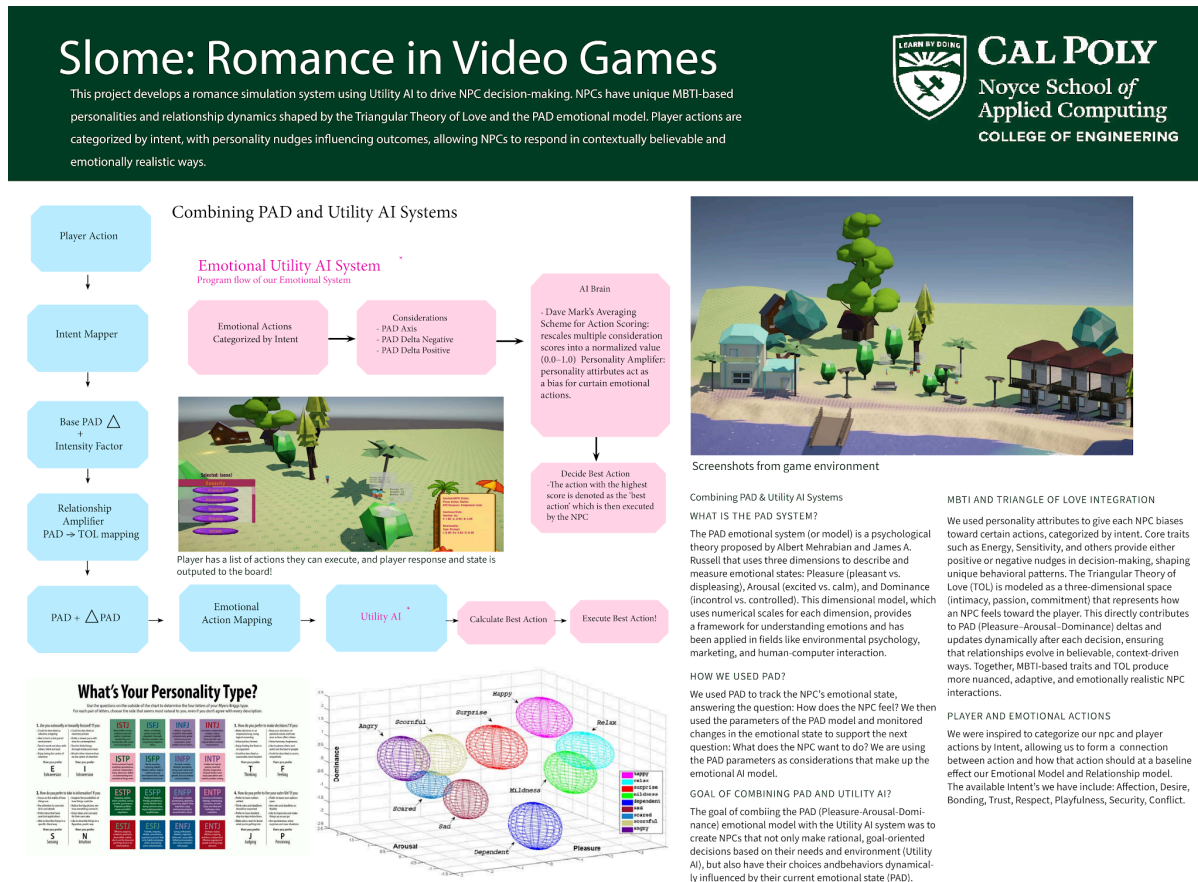


Figure 1. Uploaded project poster, page 1. It summarizes the intended player-action $\rightarrow$ intent $\rightarrow$ PAD/TOL $\rightarrow$ utility-AI workflow and the stated motivation of emotionally believable NPC responses.

1. Executive Summary

Slome is an emotional decision-making prototype for romance-oriented NPCs. Its strongest contribution is not a complete romance game, but a layered architecture that separates what happened, how the NPC feels, what the relationship means, who the NPC tends to be, and what response currently has the highest utility.

CURRENT TRUTH IN ONE SENTENCE

The snapshot successfully demonstrates data-driven emotional action selection, but relationship growth is not wired into the active scenes, personality data is inconsistent, and selected emotional actions are mostly debug/log responses rather than fully expressed gameplay behavior.

What the prototype already proves

- **A compact emotional state can drive decisions.** Pleasure, Arousal, and Dominance are exposed as numeric state and consumed by authored considerations.
- **Player actions can be normalized through intent.** Forty-six fine-grained actions map into nine reusable semantic categories, reducing authoring complexity.
- **Personality can be a bias rather than a script.** The intended architecture lets context and emotion generate the candidate score, then uses personality to nudge the result.
- **Romance can be multidimensional.** Intimacy, passion, and commitment are represented separately instead of collapsing relationship progress into one number.
- **Emotional and ordinary needs can coexist.** The NPC controller can choose between a conventional needs-based Utility AI and an emotional response brain.

What the prototype does not yet prove

- **A complete relationship arc.** The RelationshipProfile dependency is null in inspected scenes, so player actions do not currently evolve intimacy, passion, and commitment in runtime scene wiring.
- **Behavioral or conversational realism.** The selection layer chooses a response asset, but action execution is primarily logging/cooldown scaffolding; dialogue, animations, gestures, target interaction, and memory are not implemented.
- **Reliable personality differentiation.** Most personality assets contain legacy serialized fields that the current class no longer reads, and the final scene's assigned profiles do not represent cleanly authored distinct types.
- **Ethically believable romance.** Consent, boundaries, appropriateness gates, preferences, rejection, repair, and relationship-specific prerequisites are not modeled.
- **A production-ready project.** Build settings point at a missing scene, duplicate/legacy systems remain, and automated coverage is small.

Recommended immediate handoff goal

1. **Make one canonical scene run.** Fix Build Settings, choose the canonical demo scene, and remove duplicate input components.

2. **Wire relationship state correctly.** Attach a RelationshipProfile per NPC–player relationship and verify that TOL deltas persist and affect PAD amplification.
3. **Clean personality data.** Define one current schema, migrate or recreate assets, and add validation so stale profiles cannot silently default to zero.
4. **Turn decisions into observable behavior.** Implement at least one complete action in every intent category with animation/dialogue/UI feedback and clear consequences.
5. **Add regression tests and telemetry.** Test every mapping and asset, then log the full score breakdown so designers can explain why an NPC responded as it did.

Implementation dashboard

Subsystem	Status	Evidence / qualification
Intent mapping	Implemented	46 player actions map to 9 intents.
PAD state and deltas	Implemented	Runtime PAD update, clamping, last-delta storage, and 8-octant label.
Emotional Utility AI	Implemented	Consideration multiplication, compensation, utility curve, weight, personality multiplier, argmax.
Hybrid ordinary/emotional AI	Prototype	NPCController can switch between regular and emotional brains.
Triangular relationship model	Partial	Data/model code exists; active scene wiring is absent and classifier is simplified.
Personality bias	Partial	Bias code exists; most profile assets are stale/incomplete and continuous values are thresholded.
Action expression	Stub / not implemented	Selection assets execute mostly as log/cooldown; animations and dialogue are not authored.
Memory / LLM dialogue	Not implemented	Files are placeholders/comments only.
Build and test readiness	Blocker / partial	Build scene is missing; three tests cover only a small fraction of the system.

2. Project Purpose and Design Motivations

The project began from a critique of common romance mechanics: many games represent intimacy as a visible score that rises through correct gifts, repeated dialogue choices, or a fixed quest chain. Those structures are legible, but they can make relationships feel transactional and NPCs feel like reward dispensers.

2.1 The design problem

The intended NPC should not merely ask, “Did the player choose the correct romance option?” It should be able to answer a sequence of more human-readable questions:

1. **What happened?** Classify the player action and its strength.
2. **How did it affect me right now?** Update a continuous emotional state.
3. **What does it mean in this relationship?** Interpret the same act differently depending on intimacy, passion, commitment, history, and boundaries.
4. **How am I inclined to respond?** Allow personality and preferences to bias, but not dictate, behavior.
5. **What can I actually do here?** Score available actions against current state and context.
6. **How will this choice change the future?** Write relationship consequences and memories so later interactions are not isolated.

2.2 Four design commitments

Commitment	Motivation	Current mechanism
Emotional continuity	NPC state persists across interactions rather than resetting after each dialogue node.	PAD is stored numerically and updated by deltas.
Non-transactional relationships	Relationship quality should have multiple meanings and cannot be purchased through one universal currency.	Intimacy, passion, and commitment are separate.
Individual variation	Different NPCs should plausibly interpret or express the same situation differently.	Personality profiles bias response utility and PAD baselines.
Explainable emergence	Behavior should arise from data and scores that designers can inspect, not opaque random output.	Utility considerations, curves, weights, and debug logs expose decision inputs.

2.3 Why a layered simulation instead of one romance meter

A single relationship number conflates several questions. A character may trust the player deeply without romantic desire; feel intense attraction without commitment; care for the player while currently angry; or love the player but refuse a specific action. The project separates these conditions so they can coexist rather than overwrite one another.

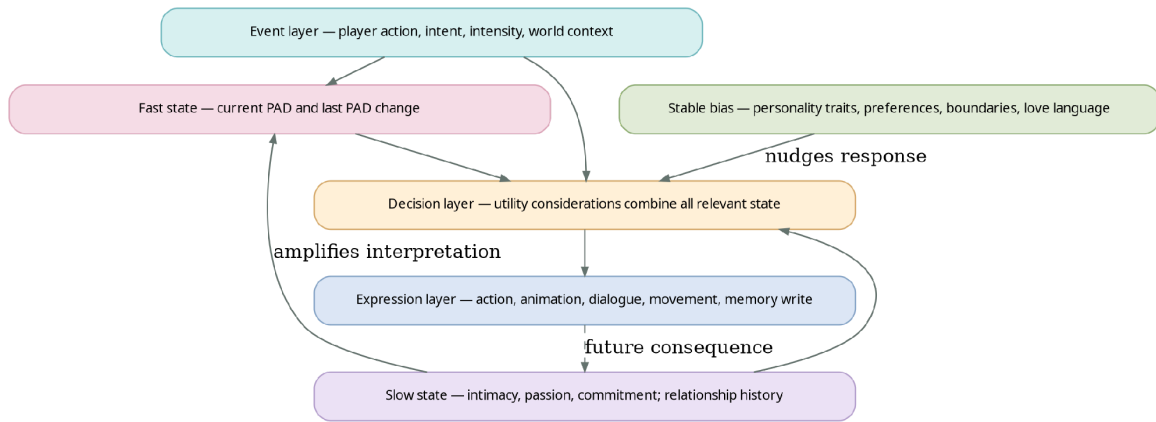


Figure 2. The intended layered model. Fast affect, slower relationship state, and stable personality contribute differently to a decision. The current code implements much of the center of this diagram; world context, boundaries, memory, and expression remain future work.

2.4 Final architectural decisions reflected in the snapshot

Decision	Why it was chosen
Use Utility AI as the selection engine	It supports many candidate responses, continuous scores, designer-authored curves, and explainable competition between actions.
Represent immediate affect with PAD	Three continuous dimensions are compact enough for runtime scoring while distinguishing affective states that a single positive/negative value cannot.
Represent relationship with Sternberg's triangle	Three dimensions allow friendship, attraction, and commitment to develop at different rates.
Translate fine actions into intent categories	Intent becomes a semantic bridge between content and simulation, preventing duplicated PAD/TOL tuning for every individual action.
Use personality as a multiplier after situational scoring	The NPC remains responsive to the situation; type nudges likely choices rather than hard-locking behavior.
Keep systems data-driven through ScriptableObjects	Actions, considerations, utility curves, and profiles can be tuned in the Unity Inspector without rewriting decision logic.
Defer generative dialogue and memory	The deterministic emotional state should become the source of truth before an LLM is allowed to verbalize or remember events.

3. Integrated Architecture

The system can be understood as a pipeline with two types of state and two types of selection. A player action updates emotional and relationship state; the emotional brain then scores response candidates. Outside a triggered interaction, the ordinary Utility AI can continue needs-based behavior.



Figure 3. Architecture reconstructed from the code snapshot. Dashed gray response expression is intentionally shown as incomplete. The relationship branch is coded but not connected in the inspected scenes.

3.1 The computational flow

- 1. Capture the player action.** The UI or input script selects one PlayerAction enum value.
- 2. Map action to intent.** IntentMapper collapses the action into Affection, Desire, Bonding, Trust, Respect, Playfulness, Security, Conflict, or Manipulation.
- 3. Calculate raw affective impact.** BasePadByIntent returns a signed $\Delta P/\Delta A/\Delta D$ vector; PlayerActionIntensity scales it.
- 4. Update relationship.** BaseRelationshipDeltaByIntent returns $\Delta \text{Intimacy}/\Delta \text{Passion}/\Delta \text{Commitment}$ and EmotionModel scales it by 0.3 before applying it to RelationshipProfile.
- 5. Amplify affect by relationship.** RelationshipAmplifier maps intimacy to Pleasure sensitivity, passion to Arousal sensitivity, and commitment to Dominance sensitivity, then clamps each action delta to ± 0.35 .
- 6. Apply emotional state.** EmotionModel stores the last delta, adds it to PAD, clamps PAD to $[0, 1]$, and classifies the current octant.
- 7. Score response actions.** Each EmotionalAction combines its considerations multiplicatively, applies Dave Mark-style compensation, evaluates a utility curve, applies weight and limited intent compatibility, then receives a personality multiplier.
- 8. Choose and execute.** EmotionBrain chooses the highest final score; NPCController transitions through Decide, Move, Execute, and Idle states.

3.2 Core equations as implemented

Runtime formulas (simplified from the current C# implementation)

```

rawDeltaPAD = BasePadByIntent(intent) * actionIntensity
deltaTOL    = BaseRelationshipDeltaByIntent(intent) * 0.3
amplifiedP  = rawP * (1 + intimacy * 0.8)
amplifiedA  = rawA * (1 + passion * 0.6)
amplifiedD  = rawD * (1 + commitment * 0.6)
deltaPAD    = clampEach(amplifiedPAD, -0.35, +0.35)
PAD         = clamp01(PAD + deltaPAD)

product     =  $\prod$  considerationScore_i
compensated = product + (1 - product) * (1 - 1/n) * product
baseUtility = UtilityCurve(compensated) * weight * intentCompatibility
finalScore  = baseUtility * personalityBias
bestAction  = argmax(finalScore)

```


DESIGN INTERPRETATION—NOT A PSYCHOLOGY CLAIM

The mapping intimacy→Pleasure, passion→Arousal, and commitment→Dominance is a project-specific computational bridge. Sternberg's and PAD theories do not prescribe this mapping. It is a plausible authoring heuristic that should be validated through playtests and tuning.

3.3 Why the combination can produce believable variation

The layers solve different problems. Intent supplies meaning; intensity supplies event magnitude; PAD supplies moment-to-moment affect; the love triangle supplies relationship interpretation; considerations convert state into action desirability; personality changes preference among otherwise plausible responses. Because the same player action enters a different state for each NPC, identical content can produce different outputs without writing a bespoke dialogue tree for every combination.

3.4 Why it is not emotionally complete yet

Believability requires more than a score. The current system does not model the cause of an emotion, the social setting, consent, expectations, prior violations, personal preferences, memory salience, habituation, or the expressive realization of the selected response. The architecture is therefore best described as an emotional action-selection foundation.

4. Utility AI and Its Design Lineage

Utility AI is the project's decision framework. Rather than following one fixed branch, each available action asks how useful it is under the current conditions. The action with the strongest score becomes the NPC's next behavior.

4.1 Where the idea comes from

Utility systems in game AI assign a normalized desirability to candidate actions based on considerations such as hunger, danger, distance, emotion, or relationship context. Dave Mark's work popularized practical scoring techniques, response curves, and compensation schemes for combining multiple considerations. The project's naming and scoring comments explicitly reference this lineage.

The ordinary NPC system also resembles the needs-based autonomy associated with The Sims: motives change over time; actions such as Eat, Work, Sleep, and DropOffResource advertise different value depending on current needs and destinations. Richard Evans's Sims 3 personality presentation provides a closely related inspiration for allowing traits to influence autonomous behavior without replacing situational decision-making.

4.2 How the regular Utility AI works

Primary code paths

Assets/Scripts/UtilityAI/AIBrain.cs
 Assets/Scripts/UtilityAI/Action.cs
 Assets/Scripts/UtilityAI/Considerations.cs
 Assets/Scripts/UtilityAI/Stats.cs
 Actions represented in the demo: Eat, Work, Sleep, DropOffResource

Each regular action has considerations tied to statistics such as hunger, energy, money, and inventory. Scores are multiplied, then the brain selects the highest action and the NPC controller moves to its destination and executes it. This establishes the project's baseline autonomy before emotional interactions are layered on top.

IMPLEMENTATION DEFECT TO FIX

The regular AIBrain uses `1 / action.considerations.Length` in the compensation calculation. Because both operands are integers, the division becomes zero for any action with two or more considerations. Use `1f / n`, as the emotional action implementation already does.

4.3 How the emotional Utility AI extends it

EmotionalAction replaces hunger and energy with emotional considerations: high/low Pleasure, Arousal, and Dominance; positive/negative change on each axis; and selected intent-match considerations. A response therefore becomes desirable because the NPC is in a particular affective state and has just experienced a particular kind of interaction.

Stage	Function in EmotionalAction	Reason
Considerations	Evaluate current PAD, last Δ PAD, and sometimes intent.	Convert state into normalized evidence for the action.
Multiplication	Treats considerations as an AND-like requirement.	One very low requirement can suppress an otherwise attractive response.
Compensation	Raises multi-consideration products to reduce excessive score collapse.	Makes authored actions with several considerations remain competitive.
Utility curve	Remaps the compensated value.	Lets designers author thresholds, urgency, saturation, or smooth ramps.
Weight	Provides a general action priority.	Supports tuning independent of the input curve.
Intent compatibility	Currently special-cases incoming Affection and Desire.	Attempts to favor contextually matching response families.
Personality multiplier	Multiplies the situational score by a small bias.	Creates individual tendency without overriding emotional context.
Argmax	EmotionBrain picks the highest score.	Deterministic and easy to debug, but can repeat.

4.4 Why Utility AI is a strong fit for romance simulation

- **It accepts continuous state.** PAD and relationship values can feed curves directly.
- **It permits competing motives.** An NPC can simultaneously want closeness, safety, distance, or playfulness, with context deciding which wins.
- **It is data-driven.** Designers can add and tune response assets without changing a central branching tree.
- **It is inspectable.** A debug view can show every consideration and multiplier, which is essential for emotionally sensitive behavior.
- **It scales compositionally.** Memory, boundaries, location, attachment, and social context can later become additional considerations.

4.5 Current limitations of the scoring approach

- **Deterministic repetition.** Pure highest-score selection has no inertia, cooldown pressure, weighted randomness, or repetition penalty.
- **Shared mutable assets.** EmotionalAction is a ScriptableObject containing runtime score and cooldown state. Two NPCs referencing the same asset can accidentally share state.
- **Uneven intent authoring.** Only a subset of response assets include an intent-match consideration, and compatibility code only explicitly handles Affection and Desire.
- **Zero-consideration default.** Hold Hands has no considerations and therefore defaults to a state-independent utility value.
- **Score normalization ends too early.** Base utility is clamped before personality; the final score can exceed 1.0 after multiplication.
- **Limited context.** No world, target, relationship-appropriateness, consent, memory, or dialogue-state considerations are in the current action assets.

5. PAD: The NPC's Immediate Emotional State

PAD—Pleasure, Arousal, and Dominance—provides the answer to “How does the NPC feel right now?” The model originated in environmental psychology through the work of Albert Mehrabian and James A. Russell and has been applied as a compact dimensional representation of affect.

5.1 What each dimension means

Axis	Conceptual range	Game-system use
Pleasure (P)	pleasant ↔ unpleasant	Valence or hedonic tone. High values indicate positive experience; low values indicate distress or dislike.
Arousal (A)	activated ↔ calm	Physiological/behavioral activation. High values indicate excitement, agitation, or alertness; low values indicate quiet or low activation.
Dominance (D)	in control ↔ controlled	Perceived control, confidence, or power. It helps distinguish affective states that have similar valence/arousal but different agency.

5.2 Why PAD was chosen

A named-emotion finite-state machine would force the project to decide in advance which emotions exist and how every interaction transitions between them. PAD instead stores a continuous state. A single vector can move gradually, be compared with thresholds, feed utility curves, and preserve mixed conditions. It is especially useful because Dominance separates high-arousal negative states such as anger and anxiety: both may be unpleasant and activated, but anger can be high-control while anxiety is low-control.

5.3 How the code uses PAD

Primary code paths

Assets/Scripts/EmotionalAI/Model/EmotionalCore.cs — PAD struct, Triangle struct, enums
 Assets/Scripts/EmotionalAI/Model/EmotionModel.cs — state update and storage
 Assets/Scripts/EmotionalAI/Model/EmotionClassifier.cs — 8-octant label
 Assets/Scripts/EmotionalAI/Intent/BasePadByIntent.cs — baseline intent deltas
 Assets/Scripts/EmotionalAI/Model/RelationshipAmplifier.cs — relationship-sensitive deltas

The current PAD state is normalized to [0, 1]. An incoming player action produces a signed delta, the delta is scaled by action intensity and relationship amplification, each component is capped at ± 0.35 , and the final PAD state is clamped to the valid range. The model also records the last delta so response actions can react not only to where the NPC is, but to the direction of the latest emotional change.

5.4 Emotion labels in the current classifier

EmotionClassifier uses a binary threshold of 0.5 on each axis, yielding eight octants. The enum names are Sad, Resigned, Anxious, Angry, Tender, ProudSecure, Awe, and Joy. These labels are useful for debugging

and display, but the utility brain primarily consumes the continuous values and deltas rather than the label itself.

P	A	D	Current label	Interpretive shorthand
Low	Low	Low	Sad	Unpleasant, quiet, low control
Low	Low	High	Resigned	Unpleasant, quiet, higher control
Low	High	Low	Anxious	Unpleasant, activated, low control
Low	High	High	Angry	Unpleasant, activated, high control
High	Low	Low	Tender	Pleasant, quiet, yielding/low control
High	Low	High	ProudSecure	Pleasant, calm, high control
High	High	Low	Awe	Pleasant, activated, low control
High	High	High	Joy	Pleasant, activated, high control

5.5 Final design interpretation

USE THE VECTOR AS THE TRUTH; TREAT LABELS AS UI

The strongest decision in the current design is that actions score against continuous PAD and Δ PAD. The octant name should remain a readable summary, not become the sole state machine. This preserves nuance and avoids hard discontinuities when one value crosses 0.5.

5.6 What PAD does not model

- **Cause and appraisal.** The vector says how the NPC feels, not whether the cause was betrayal, danger, embarrassment, jealousy, grief, or moral offense.
- **Emotion regulation.** There is no decay to baseline, suppression, coping, rumination, or habituation.
- **Mixed or discrete emotion concepts.** Three dimensions are deliberately compressed and cannot replace a richer appraisal or memory model.
- **Temporal dynamics.** Every change persists until another action changes it; the model has no time-based homeostasis.
- **Individual interpretation before the delta.** Personality currently chooses the response but does not meaningfully transform the incoming PAD delta beyond an initial baseline.

6. Triangular Theory of Love: The Relationship State

Robert Sternberg’s Triangular Theory of Love distinguishes intimacy, passion, and decision/commitment. The project uses these dimensions as a slow-changing relationship state, answering “What kind of bond does this NPC currently have with the player?”

6.1 The three relationship dimensions

Dimension	Psychological meaning	Current project mapping
Intimacy	Emotional closeness, understanding, trust, disclosure, and felt connection.	Affection, Bonding, and Trust can increase it; Conflict and Manipulation reduce it.
Passion	Attraction, desire, excitement, and motivational activation.	Desire increases it most; Playfulness and even Conflict can add activation in the current mapping.
Commitment	Decision to maintain, protect, or invest in the relationship.	Trust and Security increase it; Conflict and Manipulation reduce it.

6.2 Why this model was chosen instead of an affection bar

Separating the dimensions lets the simulation represent relationships that are not simply “more” or “less” romantic. High intimacy with low passion can support friendship or companionate bonds. High passion without commitment can support attraction or infatuation. Commitment can remain after a conflict even when current pleasure is low. This directly supports the project’s motivation to include friendship, queer relationships, varied expressions of love, and non-transactional development.

6.3 How the code uses the triangle

Primary code paths

```
Assets/Scripts/EmotionalAI/Relationship/RelationshipProfile.cs
Assets/Scripts/EmotionalAI/Intent/BaseToByIntent.cs
Assets/Scripts/EmotionalAI/Model/RelationshipAmplifier.cs
EmotionModel.ProcessPlayerAction(): relationship delta → relationship update → PAD amplification
```

Every intent has a baseline relationship delta. EmotionModel multiplies it by 0.3, applies it to RelationshipProfile, and then uses the resulting triangle to amplify the same interaction’s PAD delta. The order creates immediate feedback: a trust action can first strengthen intimacy/commitment, then those updated values can influence the emotional impact of that action.

6.4 Project-specific PAD ↔ TOL bridge

Relationship dimension	Amplified PAD axis	Gain	Design interpretation
Intimacy	Pleasure	0.8	Closeness makes relational interactions feel more emotionally positive or more painful, depending on delta sign.
Passion	Arousal	0.6	Attraction/activation increases the intensity of excitement or agitation.
Commitment	Dominance	0.6	Security and investment can affect felt control, confidence, or vulnerability.

This bridge is a deliberate game-design inference. It is valuable because it makes relationship history affect immediate emotion with only three tunable correspondences. It should not be presented as an established psychological equation.

6.5 Current implementation status and critical gap

RELATIONSHIP UPDATES ARE NOT ACTIVE IN INSPECTED SCENES

Every inspected scene instance of EmotionModel serializes relationshipProfile as null, and no RelationshipProfile component was found in the project scenes. EmotionModel therefore exits the relationship-update path. Its local triangle remains at the default 0.4/0.4/0.4 and can still amplify PAD, but the triangle does not evolve from player actions.

RelationshipProfile also attempts to label a relationship as Dislike, Stranger, Friend, CloseFriend, Crush, or Partner using threshold rules. Acquaintance exists in the enum but is never returned. The classifier is not a complete implementation of Sternberg's eight love combinations, and transition logic compares enum integer order as though all labels form one linear progression. That assumption is questionable—for example, Crush and CloseFriend are different relationship shapes, not necessarily levels on one ladder.

6.6 Recommended relationship correction

- 1. Store state per relationship pair.** A relationship belongs to NPC A toward target B, not to the NPC globally. Design now for multiple players/NPC targets.
- 2. Wire and serialize it explicitly.** Require a RelationshipProfile reference in EmotionModel and fail validation loudly if missing.
- 3. Separate dimensions from labels.** Keep I/P/C as source of truth. Derive descriptive labels for UI, not progression order.
- 4. Gate actions by appropriateness.** A kiss, confession, pressure, or intimate touch should consider consent, boundaries, existing relationship, preference, and context before applying a positive baseline.

5. **Choose update timing deliberately.** Decide whether the current action should be amplified by the relationship before or after the action changes it; document the choice.
6. **Add recovery and asymmetry.** Trust can fall quickly and recover slowly; passion, intimacy, and commitment need not share identical rates.

7. Personality Bias: MBTI / 16Personalities-Style Profiles

Personality is intended to answer “Given several plausible responses, which one is more characteristic of this NPC?” The design does not use personality as the emotional state itself. Instead, it applies a small multiplier after situational utility has been calculated.

7.1 Why MBTI-like archetypes were selected

For designers, four-letter personality labels are recognizable, concise, and easy to turn into character briefs. They provide a shared vocabulary for tendencies such as extraversion versus introversion, intuition versus sensing, feeling versus thinking, and judging versus perceiving. The project’s current data also adds an Assertive/Turbulent “Identity” axis associated with the 16Personalities framework, so it is more accurate to call the implementation MBTI-inspired or 16Personalities-style than canonical MBTI.

7.2 How the current profile is represented

Current profile and bias paths

Assets/Scripts/Personality/PersonalityTypeDefinition.cs

Continuous fields: energy, mind, nature, tactics, identity (0..1)

Additional authored fields: sensitivity, primaryLoveLanguage, five love-language values, PAD baseline

Assets/Scripts/EmotionalAI/Personality/PersonalityBiasPresets.cs

Assets/Scripts/EmotionalAI/Personality/PersonalityBias.cs

Axis	Low-side shorthand	High-side shorthand	Current evaluation
Energy	Introverted	Extraverted	Thresholded at 0.5
Mind	Sensing / Observant	Intuitive	Thresholded at 0.5
Nature	Thinking	Feeling	Thresholded at 0.5
Tactics	Prospecting	Judging	Thresholded at 0.5
Identity	Turbulent	Assertive	Thresholded at 0.5

7.3 Personality as a nudge

Each emotional response action can select a PersonalityBiasPreset. The preset references one or two axes and assigns small positive or negative factors, typically around 0.95 to 1.12. PersonalityBias combines these factors and clamps the result between 0.80 and 1.30. EmotionBrain then multiplies the response’s situational utility by this bias.

Intended decision rule

$\text{finalScore}(\text{action}, \text{npc}) = \text{situationalUtility}(\text{action}, \text{PAD}, \Delta\text{PAD}, \text{intent}) \times \text{personalityBias}(\text{action}, \text{npc})$

Design rule: personality should break ties and shape patterns—not make an emotionally impossible response win.

7.4 Why this is the correct connection to Utility AI

A type-based behavior table would make every “INTP” or “ENFJ” act alike and would override context. A post-score nudge preserves situational sensitivity: an introverted character can still confront someone when anger, dominance, and stakes are sufficiently high; an extraverted character can still withdraw when hurt or unsafe. Over many interactions, small multipliers create a recognizable distribution rather than a rigid stereotype.

7.5 Psychological caution

USE AS A FICTION-AUTHORING TOOL, NOT A DIAGNOSIS

MBTI’s categorical typing and test–retest/structural validity are debated in psychometric research. The project should not claim that a four-letter type predicts real human behavior. Keep readable archetype labels for design, but store and tune continuous traits underneath them and validate behavior through playtesting.

7.6 What the code actually contains

The snapshot includes 32 personality ScriptableObject assets, but only three visibly serialize the five current active axes and current love-language fields. Most assets retain legacy serialized names—such as expressiveness, emotional resilience, attachment style, and other fields—that no longer exist in PersonalityTypeDefinition. Unity ignores those stale names, so the current fields silently use defaults.

- **Final-scene NPC 1.** Assigned INTP-A.asset, but its serialized typeName is ENTJ-A; the current five active axes are absent/defaulted, while PAD baseline is 0.7/0.7/0.7.
- **Final-scene NPC 2.** Assigned ENFJ-A.asset, but typeName is also ENTJ-A; energy is approximately 0.006 and most current axes default to zero, with PAD baseline 0.6/0.55/0.5.
- **Thresholding loses magnitude.** PersonalityEval.IsPositive only checks whether each value is at least 0.5. A value of 0.51 behaves the same as 1.0.
- **Unused data.** Sensitivity and love-language fields are not consumed by response scoring.
- **Duplicate models.** The project contains several legacy personality representations—global PersonalityProfile, MBTIPersonalityController, a TL.Personality bool profile/NPCPersonality path, and the current PersonalityTypeDefinition.

7.7 Recommended personality direction

1. **Select one canonical schema.** Remove or quarantine legacy classes after migration.
2. **Use continuous strength.** Convert an axis value into a signed strength around 0.5 instead of a binary side.
3. **Validate assets.** An editor validator should detect missing current fields, mismatched type names, empty presets, and all-zero profiles.
4. **Separate identity from preferences.** Personality, love language, attachment strategy, boundaries, attraction, values, and communication style should be distinct data modules.
5. **Test distributions.** Run repeated scenario simulations and verify that profiles produce tendencies without deterministic stereotypes.

- 6. Offer designer labels over trait vectors.** An archetype can remain readable while the runtime consumes continuous traits with documented behavioral meanings.

8. Intent Taxonomy and Action Authoring

Intent is the semantic middle layer that makes the project scalable. Individual actions retain recognizable content names, but the simulation updates state through a smaller set of meanings.

8.1 Why intent is necessary

Without intent, each of the 46 player actions would require an independent PAD delta, TOL delta, and response-compatibility design. Intent allows related actions—such as Hug, Comfort, and Encourage—to share an affective baseline while preserving different intensity. It also gives dialogue, animation, and memory systems a compact tag for what the player was trying to do.

8.2 The nine current intents

Intent	Meaning	Mapped player actions
Affection	Warmth, care, reassurance, physical or verbal tenderness	ComplimentLooks, Hug, HoldHands, Comfort, Encourage, GiftSmall
Desire	Attraction, romantic/sexual pursuit, longing	KissQuick, KissDeep, Flirt, Seduce, LongFor
Bonding	Shared experience and mutual connection	InviteActivity, ShareStory, Reminisce, Celebrate, Support
Trust	Vulnerability, repair, reliance, promise	Apology, Confide, Forgive, AskHelp, Promise
Respect	Recognition of ability, autonomy, or worth	ComplimentSkill, Acknowledge, Admire, Defend, Praise
Playfulness	Humor, novelty, challenge, light teasing	TeasePlayful, Joke, Challenge, Surprise, PlayfulTrick
Security	Reliability, protection, steadiness	KeepPromise, Reassure, Protect, Shelter, Steady
Conflict	Opposition, criticism, withdrawal, demand	TeaseHarsh, Confront, Criticize, Withdraw, Demand
Manipulation	Coercive or instrumental influence	GiftLarge, GuiltTrip, Flatter, Pressure, Withhold

The poster lists eight intents and does not include Manipulation. The code represents the later decision: Manipulation was separated from Conflict so coercive tactics could have distinct relationship and affective consequences.

8.3 Baseline plus intensity

Each intent supplies a baseline direction in emotional and relationship space. The individual action supplies a strength multiplier. For example, both Flirt and KissDeep are Desire, but their intensities are 1.4 and 2.2 respectively. Conflict and Manipulation actions are authored at higher intensities, ranging up to 3.5, to make harmful interactions consequential.

TUNING WARNING

RelationshipAmplifier clamps each PAD component to ± 0.35 . Because several action intensities exceed 2.0, many high-intensity actions can hit the same cap and lose differentiation. The code also contains APIs/comments that call intensity “intensity01,” even though the authored range is 0.8–3.5. Normalize the naming and decide whether intensity is a 0–1 scalar or an open multiplier.

8.4 Intent as incoming meaning versus response family

The same Intent enum is used to classify the player action and to organize NPC response actions. That is useful, but the current implementation does not consistently tag every response asset with its own intent. Some assets include intent-match considerations; many Trust, Respect, Conflict, and Manipulation responses do not. GetIntentCompatibilityMultiplier only has special logic for incoming Desire and Affection. The next group should make response intent an explicit required field on EmotionalAction and implement compatibility as one centralized matrix.

8.5 Why intent alone is not consent or meaning

The author-assigned intent describes what the action category means in the system. It does not establish that the target welcomes it, believes it, or interprets it positively. A “GiftLarge” may be manipulation in this taxonomy, but even an Affection action such as a hug must be filtered through consent, boundaries, relationship context, culture, and prior history. Those appraisal layers are not yet implemented.

9. End-to-End Runtime Walkthrough

This section follows one interaction from player input to the selected NPC response and identifies the exact handoff between components.

9.1 Interaction entry points

Two input/UI paths exist. `IntentActionSelectUI` plus `NewPlayerInputManager` provides a dropdown-style selection flow: open the UI, select an intent, select an action, find a nearby NPC, and trigger the interaction. `FinalScene` appears to use an older `PlayerInputManager` path. The coexistence of both paths is part of the project's prototype history, but the next group should select one canonical input controller.

Primary input and control paths

`Assets/Scripts/Player/NewPlayerInputManager.cs`
`Assets/Scripts/UI/IntentActionSelectUI.cs`
`Assets/Scripts/Player/PlayerInputManager.cs` (legacy/final-scene path)
`Assets/Scripts/UtilityAI/NPCController.cs` (interaction target and FSM)

9.2 Processing the player action

1. **`NPCController.TriggerEmotionalInteraction(action)`**. Looks up `PlayerActionIntensity.Get(action)` and sends action plus intensity to `EmotionModel`.
2. **`EmotionModel` maps intent**. `IntentMapper.Map(action)` records the last intent.
3. **Relationship update is attempted**. A TOL delta is calculated and applied only if `relationshipProfile` is non-null.
4. **PAD delta is calculated**. $\text{BasePadByIntent} \times \text{intensity}$, then `RelationshipAmplifier` uses the triangle.
5. **State is recorded**. `lastDelta` is saved, PAD is clamped, and `EmotionClassifier` provides a debug label.
6. **The emotional trigger is set**. `NPCController`'s state machine switches to emotional decision logic when the player is nearby and the trigger is active.

9.3 Selecting the NPC response

1. **`EmotionBrain` iterates response assets**. `FinalScene` assigns 46 `EmotionalAction` `ScriptableObject`s to each of two NPCs.
2. **Each action scores its considerations**. Authored `AnimationCurves` evaluate PAD/delta/intent input.
3. **The action computes situational utility**. Scores are multiplied, compensated, curved, weighted, and adjusted by limited intent compatibility.
4. **Personality bias is applied**. The action's preset reads one or two profile axes and returns a bounded multiplier.
5. **Highest score wins**. `EmotionBrain` stores the best action and `NPCController` executes it through the FSM.

9.4 What “Execute” currently means

DECISION LAYER COMPLETE; EXPRESSION LAYER MOSTLY EMPTY

All 46 emotional action subclasses override Execute but simply call the base implementation. The base sets an optional animation trigger, updates cooldown, and writes a log. The current action assets have empty animation-trigger fields. There is no response-specific dialogue, movement, gesture, target effect, memory write, or relationship consequence.

This distinction is important for demonstrations. The prototype can explain why “Comfort,” “Withdraw,” or “Flirt” was selected, but it does not yet present that choice as a complete embodied interaction. The selected action should be treated as a semantic command for a future expression system.

9.5 Hybrid ordinary/emotional behavior

NPCController also owns the ordinary AIBrain. When there is no emotional trigger, the NPC can select needs-based actions and move toward a destination. During an emotional interaction, the controller uses EmotionBrain. This hybrid is a useful long-term architecture: romance and emotion become part of a living autonomous NPC rather than a separate dialogue-only minigame.

9.6 Timing and control notes

- **Trigger ordering.** NewPlayerInputManager calls TriggerEmotionalInteraction and sets EmotionalTriggered afterward in the same frame. It likely works on the next Update, but explicit ordering would be safer.
- **Target selection.** The new input path finds the first nearby NPC rather than presenting a target choice or deterministic nearest-target rule.
- **Duplicate components.** UI_Workspace contains NewPlayerInputManager on both EmotionalNPC1 and PlayerCommandUI, which can produce duplicated input handling.
- **Debug HUD mismatch.** The HUD can alter EmotionModel’s local triangle, but a correctly wired RelationshipProfile would become the amplifier’s source; the two controls should be unified.

10. Codebase and Asset Guide

The Unity project is organized as a research prototype rather than a clean production package. The following map identifies the current authoritative paths and the legacy areas that should not be extended without consolidation.

10.1 Project environment

Item	Snapshot value	Handoff note
Unity editor	6000.1.11f1	Open with the matching major/minor version first; allow package import to complete.
Rendering	Universal Render Pipeline 17.1	Scene visuals depend on URP.
Input	Input System 1.14	Project includes old and new player-input scripts.
Navigation	AI Navigation 2.0.8	NPCController movement uses destinations/navmesh-oriented flow.
Tests	Unity Test Framework 1.5.1	Only three tests were found.
Build Settings	Enabled Assets/Scenes/SampleScene.unity	This file is missing; fix before handoff demo/build.

10.2 Authoritative emotional-system paths

Area	Path	Responsibility
Core data	Assets/Scripts/EmotionalAI/Model/EmotionalCore.cs	PAD, Triangle, PlayerAction, Intent, EmotionOctant.
State update	Assets/Scripts/EmotionalAI/Model/EmotionModel.cs	Processes player action; updates relationship/PAD; tracks last intent and delta.
Intent mapping	Assets/Scripts/EmotionalAI/Intent/IntentMapper.cs	46-action to 9-intent switch.
PAD baseline	Assets/Scripts/EmotionalAI/Intent/BasePadByIntent.cs	Signed baseline PAD delta per intent.
TOL baseline	Assets/Scripts/EmotionalAI/Intent/BaseTolByIntent.cs	Signed relationship delta per intent.
Intensity	Assets/Scripts/EmotionalAI/Model/PlayerActionIntensity.cs	Per-player-action impact multiplier.
Amplification	Assets/Scripts/EmotionalAI/Model/RelationshipAmplifier.cs	Triangle-dependent PAD scaling and ± 0.35 clamp.
Classifier	Assets/Scripts/EmotionalAI/Model/EmotionClassifier.cs	Eight octants using 0.5 threshold.
Response action	Assets/Scripts/EmotionalAI/Actions/EmotionalAction.cs	Considerations, utility curve, weight, intent fit, cooldown/execution.

Area	Path	Responsibility
Response brain	Assets/Scripts/EmotionalAI/EmotionBrain.cs	Scores all actions, applies personality, chooses argmax.
Relationship	Assets/Scripts/EmotionalAI/Relationship/RelationshipProfile.cs	I/P/C storage and simplified type classification.
Personality bias	Assets/Scripts/EmotionalAI/Personality/*	Preset definitions, axis evaluation, and multiplier.
Controller/FSM	Assets/Scripts/UtilityAI/NPCController.cs	Coordinates regular and emotional brains, movement, execution, idle.

10.3 Data assets

- **46 EmotionalAction assets.** These are grouped by Affection, Desire, Bonding, Trust, Respect, Playfulness, Security, Conflict, and Manipulation and are assigned to demo NPC brains.
- **21 consideration assets.** Six inspect current PAD high/low, six inspect positive/negative change on each axis, and nine match incoming intent.
- **32 personality assets.** The folder appears complete by type-name count, but serialized schema quality is not complete.
- **AnimationCurve-driven tuning.** High/low axis curves and delta curves use authored nonlinear shapes; intent-match curves leave a low fallback rather than strict zero on mismatch.
- **Shared ScriptableObjects.** Treat assets as immutable definitions. Move runtime score/cooldown to per-NPC action instances or a runtime state dictionary.

10.4 Scenes

Scene	Role inferred from snapshot	Audit note
CamilaEmotionalAI	Prototype scene	Earlier emotional AI work.
UtilityAIWorkSpace	Prototype scene	Needs-based Utility AI workspace.
UtilityAIWorkSpace 1	Prototype/duplicate	Likely a copied workspace; verify before deletion.
UI_Workspace	Integration workspace	Two NPCs and newer intent/action UI; duplicate NewPlayerInputManager components detected.
DraftFinal	Draft integration	Candidate final assembly.
EmotionalAI	Prototype scene	Emotional system workspace.
ActionSystem	Prototype scene	Action-authoring/system work.
EmotionalNPC	Prototype scene	NPC emotional behavior workspace.
Camila_Final_Scene	Candidate demo	Character-specific final scene.
FinalScene	Most complete snapshot	Two NPCs; 46 emotional actions each; older input path; relationship profiles not wired.

10.5 Legacy and duplicate code to quarantine

- **Multiple personality representations.** Do not add features to all of them. Pick `PersonalityTypeDefinition` or replace it with one migrated schema.
- **Legacy `EmotionalState` in `Stats`.** The class derives from `MonoBehaviour` but is instantiated with new `EmotionalState()`, which is invalid Unity usage. Consolidate on `EmotionModel`.
- **`Memory.cs` and `MemorySystem.cs`.** These are placeholders/comments, not working memory.
- **`Relationship.cs`.** Empty placeholder.
- **`EmotionAdapter`.** A simpler entry point exists, but the actual integrated flow goes through `NPCController`; decide whether to retain it as a facade.
- **Old and new input UI.** Consolidate so there is one authoritative way to choose an action and target.

11. What Is Implemented, Partial, and Missing

The status below is based on code presence, scene serialization, and whether a feature has a visible end-to-end path. “Implemented” does not mean production-ready; it means the core runtime path is present in the snapshot.

Feature	Status	Evidence
PlayerAction and Intent enums	Implemented	46 player actions and 9 intent categories exist.
Action-to-intent mapping	Implemented	Exhaustive switch; unknown fallback returns Bonding.
Per-action intensity	Implemented	Authored multiplier for all actions.
Baseline PAD deltas	Implemented	One vector per intent.
Baseline relationship deltas	Implemented in code	One vector per intent and 0.3 scale in EmotionModel.
PAD state persistence	Implemented	State and last delta persist on component until changed.
Relationship-dependent amplification	Implemented in code	Formula exists and uses triangle values.
Relationship runtime wiring	Not implemented in inspected scenes	EmotionModel relationshipProfile is null; no component found.
Emotion label classifier	Implemented	Eight binary octants at 0.5.
Utility considerations	Implemented	PAD, ΔPAD, and intent assets with curves.
Emotional action scoring	Implemented	Multiplication, compensation, curve, weight, intent fit.
Personality multiplier	Implemented in code	Preset-based multiplier after situational score.
Personality content	Partial / invalid data	Most assets use stale schema or default values.
Ordinary needs AI	Prototype	Eat, Work, Sleep, DropOffResource path exists.
Hybrid FSM	Prototype	NPCController selects emotional or regular brain.
Emotional response content	Partial	46 semantic action assets exist.
Animation/dialogue execution	Not implemented	Assets execute base logging/cooldown; triggers empty.
Relationship labels	Partial	Simplified thresholds; Acquaintance unused; linear ordering assumption.
Love languages	Data only	Fields exist but do not affect response or interpretation.
Attachment styles	Not implemented / legacy	Appears in stale assets/conversation intent, not current active logic.
Memory/history	Not implemented	Placeholder comments/files only.
LLM/NLG dialogue	Not implemented	No runtime model or prompt integration.

Feature	Status	Evidence
Consent/boundaries	Not implemented	No gates or preference/appraisal checks.
Save/persistence	Not implemented	No durable relationship/emotion storage.
Automated tests	Partial	Three tests; major paths uncovered.
Buildable canonical demo	Blocker	Build Settings references missing SampleScene.

11.1 Important distinction: prototype breadth versus vertical completeness

The project has broad conceptual coverage: needs, emotion, relationship, personality, actions, UI, scene prototypes, and future memory ideas. Its main weakness is vertical completeness. Few interactions travel from authored player action through validated state, appropriate selection, visible response, relationship consequence, memory, and persistence. The next group should prioritize one complete vertical slice rather than adding more parallel frameworks.

12. Technical Audit, Risks, and Design Debt

The following issues are prioritized by whether they block the current design from behaving as described, create silent false results, or make future extension dangerous.

12.1 P0 — correctness and demo blockers

Issue	Impact	Required correction
Broken Build Settings	Enabled scene path is Assets/Scenes/SampleScene.unity, but the scene is absent.	Select a canonical scene, add it to Build Settings, and make a clean build before any feature work.
RelationshipProfile not wired	TOL state does not update in inspected scenes.	Add/serialize a valid per-target relationship component and create an automated scene validation.
Stale personality assets	Profiles silently default active fields, defeating NPC differentiation.	Recreate/migrate profiles and block play mode when required fields are missing.
No visible emotional execution	The demo reports selected action but does not embody it.	Implement a minimal expression adapter and complete one action per intent.
Regular utility compensation bug	Integer division invalidates intended multi-consideration adjustment.	Change $1 / n$ to $1f / n$ and add unit tests.

12.2 P1 — architecture risks

Risk	Why it matters	Recommended pattern
Shared mutable ScriptableObjects	Score/cooldown can leak between NPCs that reference the same action asset.	Keep definitions immutable; create per-agent runtime action state.
Multiple competing state models	Emotion/personality logic can diverge and developers may wire the wrong class.	Create one namespace/module and deprecate legacy files.
Relationship is NPC-global	Cannot model different feelings toward multiple targets.	Key relationship state by source and target IDs.
Inconsistent intent compatibility	Some response families are context-aware; others are not.	Require responseIntent and use one compatibility/appraisal matrix.
Intensity saturation	High multipliers hit ± 0.35 , erasing distinctions.	Plot/tune resulting deltas and choose a coherent scale.
No temporal regulation	PAD accumulates permanently and can pin at 0 or 1.	Add baseline return, decay, event half-life, or regulation policy.
Linear relationship enum ordering	Compares qualitatively different relationship shapes as upgrades/downgrades.	Use labels as categorical descriptions and compare dimensions directly.
Unknown action fallback to Bonding	A missing mapping silently becomes positive/benign.	Throw/log an error or use Intent.Unknown with zero impact.

12.3 P2 — content and behavioral quality

- **Action naming and metadata.** Correct Surprise, Critize, Acknowledge, and the Act_Deflect/Act_QuietPresence mismatch. Fix mislabeled consideration display names.
- **Action completeness.** Hold Hands has no considerations; Flirt has no personality bias preset; audit all 46 assets with a validator.
- **No prerequisites or refusal.** A high-intimacy action can be selected or interpreted positively without relationship/consent checks.
- **No diversity of expression.** One semantic action should have multiple animation/dialogue realizations conditioned on mood, personality, culture, and context.
- **No anti-repetition logic.** Add cooldowns, recent-action penalties, hysteresis, or weighted selection among near-best actions.
- **No explanation UI.** Designers need a score breakdown that shows each consideration, curve output, compensation, weight, intent multiplier, personality multiplier, and final rank.

12.4 Psychological and ethical design risks

- **Typology determinism.** Do not encode harmful stereotypes or imply that all people of a type share the same response.
- **Positive-by-category assumptions.** Affection and Desire are not inherently welcome; target boundaries and context determine valence.
- **Manipulation as effective strategy.** The simulation should represent harm, loss of trust, detection, and boundaries rather than reward coercion as a romance optimization path.
- **One-sided relationship state.** Realistic relationships may require each actor's separate state and explicit consent; mutual labels should be derived only when appropriate.
- **Mechanics transparency.** Avoid turning the triangle into another visible optimization meter that encourages "farming" intimacy, passion, or commitment.
- **LLM overreach.** A language model should not invent relationship facts, consent, or state changes outside the deterministic simulation.

13. Prioritized Future-Work Roadmap

The roadmap favors correctness and vertical integration before adding new psychological frameworks. Each phase should end with a playable, testable improvement rather than only new data structures.

Phase 0 — Stabilize the snapshot (1–3 days)

Deliverable	Definition of done
Canonical demo scene	One selected scene opens without missing references, contains player + at least two NPCs, and is the only enabled build scene.
Build succeeds	Desktop development build launches and reproduces the editor interaction.
Dependency validation	Editor script reports missing EmotionModel, RelationshipProfile, personality, brain, action list, animator, and UI references.
Code baseline	Repository tag/branch created; legacy folders labeled; README records Unity version and startup steps.
Known-issue test	Regular AIBrain compensation fixed and covered by a unit test.

Phase 1 — Complete one emotional vertical slice (1–2 weeks)

Workstream	Required outcome
Relationship wiring	Per-NPC→player RelationshipState updates I/P/C and survives multiple interactions.
One action per intent	At least nine responses have actual animation/dialogue/UI realization and target-aware execution.
Score inspector	Runtime panel shows every candidate, each consideration, personality factor, and why the winner won.
Asset validation	All 46 actions have response intent, at least one consideration or explicit constant rationale, a valid bias preset, and clean names.
Tests	Coverage for all 46 action mappings/intensities, 9 PAD/TOL baselines, amplifier, classifier, and scoring math.
State reset/decay	A simple configurable PAD return-to-baseline prevents permanent saturation.

Phase 2 — Make relationships believable (2–4 weeks)

Feature	Design requirement
Appraisal and appropriateness	Compute whether an act is welcomed, expected, violating, public/private, sincere, and consistent with history before assigning its emotional sign.
Consent and boundaries	Explicit target preferences and boundaries gate physical/romantic actions; rejection is a valid response and updates state appropriately.
Relationship asymmetry	Each actor stores their own I/P/C toward the other; mutual status is derived rather than assumed.
Repair and rupture	Apology, forgiveness, promises, and violations use memory/context, not unconditional positive deltas.
Nonlinear growth	Use diminishing returns, thresholds, trust loss asymmetry, and relationship-specific change rates.
Relationship labels	Derive descriptive Sternberg-like combinations from I/P/C without treating them as a single ordered ladder.

Phase 3 — Repair personality and add individual difference (2–3 weeks)

Feature	Design requirement
Canonical trait schema	Continuous runtime traits with documented meanings; readable archetype label is optional metadata.
Profile migration	Recreate all profile assets and verify the label, axis values, PAD baseline, preferences, and response tendencies.
Continuous biases	Bias strength reflects distance from midpoint and can combine more than binary axis sides.
Love language	Model preferred ways of expressing and receiving care as separate perceptions—not a reward multiplier.
Attachment strategy	Add dynamic relationship strategies (seeking reassurance, distancing, secure repair) as needs/appraisals, not fixed stereotypes.
Behavioral validation	Scenario suite compares action distributions across profiles and checks that situation remains stronger than type.

Phase 4 — Memory, context, and dialogue (3–6 weeks)

Feature	Design requirement
Event memory	Record actor, target, action, intent, appraisal, pre/post PAD, pre/post relationship, location, witnesses, and salience.

Feature	Design requirement
Short/long-term memory	Recent events influence immediate choices; salient summaries persist and decay/merge over time.
World context	Location, privacy, activity, danger, social norms, and availability become utility considerations.
Dialogue realization	Selected semantic action plus state/personality/memory is converted into a line or dialogue beat.
LLM guardrails	LLM receives structured state and can only realize a selected action; deterministic code owns state changes and validates outputs.
Continuity tests	Generated dialogue must not contradict known relationship, boundaries, names, memories, or current intent.

Phase 5 — Production quality and research evaluation

- **Telemetry and tuning.** Capture state transitions, score distributions, repetitions, rejection rates, and relationship trajectories.
- **Designer tooling.** Provide curve previews, scenario simulation, asset validation, batch comparison, and relationship trajectory visualization.
- **Player-facing opacity.** Decide which emotions are communicated through performance rather than exposed as meters.
- **Accessibility and inclusion.** Support varied relationship structures, aromantic/asexual identities, queer pairings, cultural communication styles, and non-romantic intimacy without treating one path as default.
- **User study.** Compare perceived believability, agency, consent awareness, predictability, and emotional continuity against a simple relationship-point baseline.
- **Save migration.** Version emotional/relationship data so future schema changes do not corrupt saves.

14. Recommended Target Architecture

The current prototype should be evolved by preserving its strongest idea—the compositional Utility AI pipeline—while enforcing clear ownership of state, immutable definitions, and explicit appraisal before state changes.

14.1 Separate definition, runtime state, and services

Layer	Contents	Ownership / rule
Definitions (immutable ScriptableObjects)	PlayerActionDefinition, ResponseActionDefinition, ConsiderationDefinition, PersonalityArchetype, animation/dialogue realization sets.	Designer-authored data only; never store per-NPC score or cooldown.
Agent runtime state	AffectState, NeedsState, PersonalityTraits, Preferences, Boundaries.	Owned by one NPC instance.
Relationship runtime state	RelationshipKey(source,target) → I/P/C, trust facets, attraction, commitments, memories.	Owned per directed pair; persisted.
Interaction context	Actor, target, location, witnesses, current activity, action, consent signal, history.	Created per event and passed through appraisal.
Appraisal service	Determines interpreted intent, welcome/violation, expectedness, sincerity, control, relevance, and signed deltas.	Replaces unconditional positive intent assumptions.
Decision service	Scores response definitions from state + context and returns ranked candidates with explanation.	Pure/testable; no side effects.
Execution service	Resolves selected action into animation, dialogue, navigation, VFX, and consequences.	Observable behavior and state writes.
Memory service	Writes structured event and retrieves relevant summaries.	Supports continuity and future dialogue.

14.2 Proposed interaction sequence

- 1. Create InteractionContext.** Capture source, target, action definition, strength, location, witnesses, and previous state.
- 2. Run target appraisal.** Convert authored action intent into target-perceived meaning using preferences, boundaries, relationship, memory, and personality.
- 3. Compute deltas.** Apply appraisal-sensitive PAD and relationship changes; avoid changing state for impossible/unobserved events.
- 4. Update fast state.** Apply PAD impulse and schedule decay/regulation toward baseline.
- 5. Rank responses.** Utility AI reads current state, relationship, context, recent actions, and available realizations.
- 6. Select with controlled variation.** Choose among near-top candidates using hysteresis, temperature, or weighted top-k while respecting hard gates.
- 7. Execute semantic response.** Animation/dialogue system realizes the chosen action.

8. **Commit consequences and memory.** Write relationship changes caused by the response, not just by the initiating event, and persist the event summary.

14.3 Recommended scoring decomposition

Suggested decomposition

eligible(action) = hard prerequisites, consent, capability, target, cooldown
 appraisal = interpret(event, relationship, memory, boundaries, preferences)
 situational = combine(PAD, Δ PAD, needs, relationship, context, recent history)
 character = personality + communication style + values + attachment strategy
 quality = anti-repetition + animation/dialogue availability + narrative constraints
 finalScore = situational $\times$ character $\times$ quality
 selection = weighted choice among candidates within ϵ of best, or deterministic in debug mode

14.4 LLM role, when added

THE LLM SHOULD VERBALIZE—NOT DECIDE THE EMOTIONAL TRUTH

Pass the selected semantic response, current affect, relationship dimensions, relevant memories, personality voice, boundaries, and scene context to the language model. Validate that the output remains within the selected intent and does not invent consent, memories, state changes, or relationship facts.

This keeps the simulation reproducible and testable. It also makes failures recoverable: a generated line can be replaced or regenerated without changing the underlying relationship state or decision.

15. Incoming-Team Runbook

The following sequence is designed to get a new group from repository checkout to a trustworthy baseline without accidentally extending legacy paths.

15.1 First-day setup

1. **Archive the uploaded snapshot.** Keep the original ZIP unchanged and create a new source-control repository or migration branch.
2. **Install Unity 6000.1.11f1.** Open the project from the folder containing Assets, Packages, and ProjectSettings.
3. **Allow packages to resolve.** Confirm URP, Input System, AI Navigation, and Test Framework compile.
4. **Inspect Console before scene changes.** Record compile warnings/errors and missing serialized references.
5. **Choose a canonical scene.** Start by comparing FinalScene and UI_Workspace; select one and document why.
6. **Fix Build Settings.** Remove missing SampleScene and add the canonical scene.
7. **Create a clean build.** Do not proceed until editor and build behavior match.

15.2 Minimum scene wiring checklist

Object/area	Required wiring
Player object	One input component, interaction radius/target rule, UI reference.
NPC object	NPCController, EmotionModel, EmotionBrain, regular AIBrain (if used), animator/nav agent.
Relationship	Directed RelationshipState/Profile linked to EmotionModel and correct target.
Personality	Validated current profile with all active fields and matching label.
Response actions	Unique immutable definitions; all required intents assigned; no missing considerations.
Destinations/needs	Eat/Work/Sleep/Drop-off destinations wired if ordinary autonomy is included.
UI/debug	Action selector, selected-target display, PAD/TOL display, score inspector.
Persistence/reset	Clear test reset and, later, save/load path.

15.3 Baseline acceptance scenarios

Scenario	Expected evidence
Affection vs conflict	From the same baseline, Hug should increase Pleasure while Criticize decreases it; last-delta considerations should change response rankings.
Relationship amplification	Run the same action at low and high intimacy/passion/commitment and verify expected PAD component changes.
Personality nudge	Two profiles in identical state should show modest score differences, but a strong situational change should be able to reverse the winner.
Consent gate	An unwelcome kiss should not use the same positive Desire appraisal as a welcomed kiss.
Decay	After no new events, PAD should move predictably toward the profile baseline.
Multi-NPC isolation	Two NPCs sharing action definitions must have independent cooldowns, scores, PAD, and relationships.
Repeat avoidance	Repeated identical events should not force the same response indefinitely when alternatives are close.
Save/load	State before save equals state after load within defined precision.

15.4 Handoff questions the new group should decide explicitly

- **What is the canonical player experience?** Free-roaming systemic interactions, authored dialogue scenes, or a hybrid?
- **What is visible to the player?** Raw PAD/TOL meters, qualitative body language, relationship labels, or no explicit numbers?
- **What is the unit of time?** Per-event impulses only, real-time decay, daily cycles, or narrative turns?
- **How is consent represented?** Explicit UI, prior agreements, contextual signals, or a combination?
- **What relationships are supported?** Friendship, romance, queer/ace/aro paths, polyamory, chosen family, rivalry, and breakup/repair?
- **How deterministic should behavior be?** Strictly reproducible for research/debug, or varied within plausible bounds for play?
- **What is the evaluation claim?** A technical prototype, a design framework, or evidence of increased perceived emotional realism?

15.5 Definition of a successful next milestone

VERTICAL SLICE MILESTONE

A player performs one of at least nine intent-diverse actions on either of two genuinely different NPCs. Each NPC appraises the event using relationship and boundaries, updates PAD and directed TOL state, selects a traceable response through Utility AI, expresses it through animation/dialogue, records a memory, and produces a reproducible score explanation. The scene builds and the scenario is covered by automated tests.

Appendix A. Intent Tuning Values

These are the current authored baselines from BasePadByIntent.cs and BaseTolByIntent.cs. Relationship deltas are multiplied by 0.3 at runtime before they are applied. PAD deltas are multiplied by the action-specific intensity, then relationship-amplified and clamped per axis to ± 0.35 .

Intent	ΔP	ΔA	ΔD	$\Delta \text{Intimacy}$	$\Delta \text{Passion}$	$\Delta \text{Commitment}$
Affection	+0.20	+0.05	-0.05	+0.10	+0.02	+0.05
Desire	+0.18	+0.18	+0.06	+0.05	+0.12	+0.03
Bonding	+0.15	+0.10	0.00	+0.08	+0.04	+0.08
Trust	+0.10	-0.05	-0.10	+0.12	+0.01	+0.12
Respect	+0.15	0.00	+0.15	+0.03	+0.01	+0.07
Playfulness	+0.15	+0.15	0.00	+0.04	+0.07	+0.02
Security	+0.10	-0.05	+0.15	+0.02	0.00	+0.12
Conflict	-0.20	+0.15	+0.15	-0.10	+0.05	-0.08
Manipulation	-0.15	+0.05	-0.10	-0.08	+0.03	-0.10

Interpretation notes

- **Trust lowers arousal and dominance in the baseline.** This can represent calming vulnerability/yielding, but may not fit every trust event; appraisal should eventually differentiate apology, confiding, asking for help, and promising.
- **Conflict raises arousal and dominance while lowering pleasure.** This naturally supports an angry high-control state, but withdrawal may instead lower dominance/arousal; action-specific appraisal would improve this.
- **Manipulation lowers intimacy and commitment but slightly raises passion.** This may represent activation or attraction pressure, but risks suggesting coercion produces romantic passion; validate or replace with a separate activation dimension/context.
- **All positive-category deltas are unconditional.** Future appraisal must be able to invert, reduce, or reject them based on target interpretation and consent.

Current action-intensity range

Category	Range	Examples
Affection	0.9–1.5	GiftSmall 0.9; HoldHands 1.0; Hug 1.5
Desire	1.2–2.2	LongFor 1.2; Flirt 1.4; KissDeep 2.2
Bonding	1.0–1.6	Reminisce 1.0; Support 1.3; Celebrate 1.6
Trust	1.0–1.6	AskHelp 1.0; Confide 1.3; Forgive 1.6

Category	Range	Examples
Respect	1.0–1.8	Acknowledge 1.0; Praise 1.2; Defend 1.8
Playfulness	0.8–1.3	Joke 0.8; TeasePlayful 0.9; Surprise 1.3
Security	1.0–1.8	Steady 1.0; Reassure 1.3; Protect 1.8
Conflict	2.2–3.0	Withdraw 2.2; Criticize 2.8; Confront 3.0
Manipulation	2.0–3.5	Flatter 2.0; GiftLarge 3.2; GuiltTrip 3.5

Appendix B. Complete Player-Action Taxonomy

The exact mapping below comes from IntentMapper.cs. It is useful for content migration, test generation, and checking UI completeness.

Intent	Player action
Affection	ComplimentLooks
Affection	Hug
Affection	HoldHands
Affection	Comfort
Affection	Encourage
Affection	GiftSmall
Desire	KissQuick
Desire	KissDeep
Desire	Flirt
Desire	Seduce
Desire	LongFor
Bonding	InviteActivity
Bonding	ShareStory
Bonding	Reminisce
Bonding	Celebrate
Bonding	Support
Trust	Apology
Trust	Confide
Trust	Forgive
Trust	AskHelp
Trust	Promise
Respect	ComplimentSkill
Respect	Acknowledge
Respect	Admire
Respect	Defend
Respect	Praise

Intent	Player action
Playfulness	TeasePlayful
Playfulness	Joke
Playfulness	Challenge
Playfulness	Surprise
Playfulness	PlayfulTrick
Security	KeepPromise
Security	Reassure
Security	Protect
Security	Shelter
Security	Steady
Conflict	TeaseHarsh
Conflict	Confront
Conflict	Criticize
Conflict	Withdraw
Conflict	Demand
Manipulation	GiftLarge
Manipulation	GuiltTrip
Manipulation	Flatter
Manipulation	Pressure
Manipulation	Withhold

Appendix C. Validation and Test Plan

The current project contains three tests: a RelationshipAmplifier clamp test, partial emotion-octant reachability checks, and a player-action PAD/intent/delta test. The following suite would make the system safe to extend.

Test area	Required assertions
Mapping completeness	Every PlayerAction maps exactly once; no fallback used; every Intent has at least one action.
Intensity completeness	Every PlayerAction has a documented finite positive intensity in the intended scale.
PAD/TOL tables	Nine expected vectors match source data and sign conventions.
Amplifier	Zero/high triangle values, negative deltas, axis gains, and ± 0.35 caps.
PAD state	Clamp, lastDelta, baseline initialization, reset, and decay.
Classifier	All eight octants and exact behavior at 0.5.
Utility math	$n=0/1$ /many considerations, compensation, curve, weight, clamp, and personality.
Intent compatibility	Every incoming/response intent pair has expected multiplier or gate.
Personality	Continuous strengths, midpoint, multi-axis combination, clamp, and profile validation.
Relationship	Directed pair isolation, update timing, label derivation, and persistence.
Shared assets	Two NPCs using one definition never share cooldown or score state.
FSM	Regular/emotional transitions, target loss, interruption, movement failure, and cooldown.
UI	Every intent/action appears once; target choice is deterministic; input component is unique.
Scene validation	Required references and canonical build scene.
Vertical integration	Action $\rightarrow$ appraisal $\rightarrow$ PAD/TOL $\rightarrow$ score $\rightarrow$ response $\rightarrow$ expression $\rightarrow$ memory.
Ethical gates	Unwelcome/intolerable actions cannot receive default-positive appraisal or forced acceptance.

Recommended telemetry record

One trace record should make every decision reproducible

eventId, time, actorId, targetId, playerAction, authoredIntent, perceivedIntent
actionIntensity, consent/appraisal flags, prePAD, rawDeltaPAD, amplifiedDeltaPAD, postPAD

preTOL, deltaTOL, postTOL, relationshipLabel
candidateAction, considerationInputs, curveOutputs, product, compensation, weight
intentCompatibility, personalityBias, repetition/cooldown factor, finalScore, selected
expressionId, dialogueId, animationId, memoryId

Appendix D. Glossary

Term	Meaning in this project
Affect	A broad term for emotional feeling/state. In this project, represented primarily through PAD.
Appraisal	The interpretation of an event in relation to goals, expectations, control, responsibility, and relevance. Not implemented yet.
Consideration	A normalized input to a Utility AI action, usually evaluated through an AnimationCurve.
Dominance	PAD dimension representing felt control or power.
EmotionalAction	ScriptableObject definition for an NPC response candidate.
EmotionBrain	Component that scores EmotionalAction candidates and selects the highest final score.
Intent	Semantic category assigned to an initiating player action; current code has nine.
PAD	Pleasure–Arousal–Dominance emotional vector.
Personality bias	Small score multiplier based on the NPC profile and the action's bias preset.
Response curve	Function that maps a raw normalized consideration or combined score to utility.
ScriptableObject	Unity data asset type used for action, consideration, and personality definitions.
TOL / Triangle	Project shorthand for Sternberg-style Intimacy, Passion, and Commitment.
Utility AI	Decision approach that assigns desirability scores to candidate actions and selects one.
Vertical slice	One complete end-to-end interaction with all required systems connected and visible.

Appendix E. Sources and References

The codebase and poster are the primary sources for implementation claims. The external works below explain the theories and design lineage. The report deliberately distinguishes their original claims from the project's own computational mappings.

Source role	Reference
Project artifact	Slome: Romance in Video Games. Cal Poly project poster, page 1. Uploaded source reviewed 5 August 2026.
Project code	slome-main Unity project snapshot. Unity 6000.1.11f1. Uploaded source reviewed 5 August 2026.
PAD foundation	Mehrabian, A., & Russell, J. A. (1974). <i>An Approach to Environmental Psychology</i> . MIT Press.
PAD framework	Mehrabian, A. (1996). Pleasure-arousal-dominance: A general framework for describing and measuring individual differences in temperament. <i>Current Psychology</i> , 14, 261–292.
Love theory	Sternberg, R. J. (1986). A triangular theory of love. <i>Psychological Review</i> , 93(2), 119–135.
Utility AI	Dill, K. (2013). An introduction to utility theory. In S. Rabin (Ed.), <i>Game AI Pro</i> .
Utility scoring	Mark, D. (2009). <i>Behavioral Mathematics for Game AI</i> . Charles River Media.
The Sims traits	Evans, R. (2010). Modeling individual personalities in The Sims 3. Game Developers Conference presentation.
Needs-based AI	Zubek, R. Needs-Based AI. Draft/technical discussion of motive-based autonomous agents.
The Sims objects	Programming objects in The Sims. Technical notes describing object interaction and motive advertisements.
MBTI caution	Myers & Briggs Foundation / Journal of Psychological Type. Review and psychometric synthesis materials on the MBTI; use alongside independent psychometric literature.
Future memory	Park, J. S., et al. (2023). Generative Agents: Interactive Simulacra of Human Behavior. <i>Proceedings of UIST 2023</i> / arXiv preprint.

Final handoff principle

PRESERVE THE ARCHITECTURE; IMPROVE THE CLAIMS AND CONNECTIONS

The project's durable contribution is the separation of affect, relationship, personality, and decision utility. The next group should keep those layers explicit, make every dependency real in the scene, add appraisal/consent/memory before claiming emotional realism, and treat psychological models as design abstractions to be tested—not as deterministic simulations of people.